GEOMETRIC METHODS FOR THE ANALYSIS OF FAST–SLOW DYNAMICAL SYSTEMS

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Abstract: In this project, we introduce fast–slow systems and analyze their dynamics within a general theoretical framework. We discuss the concept of the critical manifold and study its influence on the system's dynamics. Fenichel's Theorem is presented, showing how it is used to analyze normally hyperbolic submanifolds of the critical region, and the blow-up method is introduced to study non-hyperbolic equilibrium points. We provide a geometric description of the relaxation oscillations that occur in fast–slow systems and prove their existence. Finally, we discuss canard trajectories and the phenomenon of canard explosion.

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1 INTRODUCTION

Fast–slow dynamical systems arise naturally in a wide variety of applications, including biology, chemistry, neuroscience, and engineering, where processes evolve on widely separated time scales. Such systems typically exhibit rich and intricate dynamics that cannot be captured by classical perturbation techniques. In particular, the presence of multiple time scales often leads to non-hyperbolic behavior, loss of normal hyperbolicity, and the emergence of complex phenomena such as relaxation oscillations and canard explosions.

A standard framework for the analysis of fast–slow systems is provided by geometric singular perturbation theory (GSPT), which allows one to decompose the dynamics into fast and slow subsystems. While GSPT offers powerful tools for understanding dynamics near normally hyperbolic invariant manifolds, it breaks down near critical points where hyperbolicity is lost. Examples of such points include folds, canards points and other degenerate singularities, which play a crucial role in the onset of oscillatory behavior and canard phenomena.

The blow-up method has emerged as a fundamental technique for resolving non-hyperbolic singularities in fast–slow systems. By replacing a singular point with a higher-dimensional manifold, the blow-up method transforms degenerate dynamics into a collection of regular vector fields defined on suitable charts. This approach enables a detailed local and global analysis of the flow near critical regions, restoring hyperbolicity and allowing classical dynamical systems techniques to be applied. In the context of fast–slow systems, the blow-up method has proven particularly effective in the study of relaxation oscillations, canard explosions, and the transition between different dynamical regimes.

The goal of this work is to present a systematic introduction to fast–slow systems and the blow-up method, and to address selected results from the work of Krupa and Szmolyan [6]. We begin by reviewing the basic theory of fast–slow systems, including critical manifolds, reduced and layer problems, and the role of normal hyperbolicity. We then introduce the blow-up method for fast–slow systems and explain how quasihomogeneous blow-ups can be used to desingularize non-hyperbolic points.

These theoretical tools are subsequently applied to the study of relaxation oscillations, highlighting the geometric mechanisms underlying their formation. Finally, we focus on the canard explosion phenomenon, where solutions undergo a rapid transition between small and large amplitude oscillations.

This work aims to provide both a coherent theoretical framework and concrete examples, serving as an accessible introduction to modern geometric methods for the analysis of fast–slow dynamical systems.

2 FAST-SLOW SYSTEMS

In this section, we introduce the so-called fast–slow systems. Our aim is to present the basic definitions and establish a clear notation that will be used consistently in the remainder of the text.

Definition 2.1. For $m, n \in \mathbb{N}$, an (m, n)–**fast-slow system** is a system of $m+n$ ordinary differential equations of the form:

$$\varepsilon \frac{dx}{d\tau} = \varepsilon \dot{x} = f(x, y, \varepsilon), \qquad \frac{dy}{d\tau} = \dot{y} = g(x, y, \varepsilon), \tag{2.1}$$

where $f : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^m$ and $g : \mathbb{R}^m \times \mathbb{R}^n \times \mathbb{R} \to \mathbb{R}^n$ are smooth functions (at least, C^2), and $0 < \varepsilon \ll 1$.

Notice that considering $t = \tau/\varepsilon$, system (2.1) is equivalent to:

$$\frac{dx}{dt} = \dot{x} = f(x, y, \varepsilon), \qquad \frac{dy}{dt} = \dot{y} = \varepsilon g(x, y, \varepsilon). \tag{2.2}$$

The fast time scale is represented by t, while τ denotes the slow time scale. Although it is common in the literature to use prime notation (e.g. x' or y') instead of dot notation (e.g. $\dot{x}$ or $\dot{y}$), we will adopt the latter throughout this work for consistency.

Note that on the slow time scale, the dynamics of x is fast: its velocity is proportional to $1/\varepsilon$, and since ε is small, this velocity is large. Conversely, on the fast time scale, the dynamics of y is slow: its velocity is proportional to ε, and since ε is small, this velocity is small. For this reason, we refer to $x \in \mathbb{R}^m$ as the fast variable(s) and $y \in \mathbb{R}^n$ as the slow variable(s).

As mentioned above, these systems are dynamical systems in which some variables evolve on a much faster time scale than others. Such behavior is common in applications, and many real-world models can be cast in this framework. Well-known examples include the FitzHugh–Nagumo model from neuroscience, the Van der Pol oscillator, which arises in neural bursting and the modeling of electrical circuits, as well as various models in climate dynamics or aerodynamics.

One could ask how the model behaves when we consider the limiting case $\varepsilon = 0$.

Definition 2.2. Given $\varepsilon = 0$, the system (2.1) in the slow time scale becomes a system of differential-algebraic equations:

$$0 = f(x, y, 0), \qquad \frac{dy}{d\tau} = g(x, y, 0). \tag{2.3}$$

This system is called the **slow subsystem**; its equations are referred to as the reduced equations, and its flow as the slow (or reduced) flow.

Definition 2.3. For $\varepsilon = 0$, system (2.2) in the fast time scale is called the **fast subsystem**:

$$\frac{dx}{dt} = f(x, y, 0), \qquad \frac{dy}{dt} = 0. \tag{2.4}$$

The set of equations in system (2.4) is referred to as the layer equations, and its flow as the fast flow.

Depending on the region of phase space and the variable under consideration, the flow may evolve either relatively rapidly or slowly. The two subsystems enable the analysis of the dynamics across distinct time scales.

Definition 2.4. The **critical manifold** of the fast subsystem (2.4) is the set:

$$C_0 = \{(x, y) \in \mathbb{R}^m \times \mathbb{R}^n : f(x, y, 0) = 0\} \tag{2.5}$$

Observation 2.5. The critical manifold is sometimes referred to as the slow manifold and denoted by S_0 because the slow flow is defined on it.

Example 2.6. As an example, we consider the **FitzHugh-Nagumo model**:

$$\frac{dv}{dt} = -v(v-1)(v-a) - w + I, \tag{2.6}$$

$$\frac{dw}{dt} = \varepsilon(v - \gamma w), \tag{2.7}$$

where v and w are real variables, and a, I and $\gamma \in \mathbb{R}$ are parameters, is an example of fast–slow system. Precisely, is a fast–slow system in the fast time scale t, with v the fast variable and w the slow variable.

The critical manifold of the system is

$$C_0 = \{(v, w) \in \mathbb{R}^2 : w = -v^3 + (a+1)v^2 - av + I\},$$

which is a cubic–shaped curve. If we consider the slow subsystem, the motion around C_0 will be governed by the equation $\frac{dw}{d\tau} = v - \gamma w$ in the slow time scale τ.

To further analyze critical manifolds, we need to recall first the so called Hartman–Grobman theorem for flows.

Theorem 2.7. (**Hartman–Grobman theorem for flows**). Let $U \subseteq \mathbb{R}^n$ be an open set and let $X : U \to \mathbb{R}^n$ be a C^1 vector field having a hyperbolic equilibrium point x^*. Let $\varphi_t : U \to \mathbb{R}^n$ denote its flow.

Then, there exists a neighborhood $N \subseteq U$ of x^* on which φ_t is topologically conjugate to the linear flow

$$\psi_t(x) = e^{DX(x^*)t}x.$$

Corollary 2.8. (Using conjugacy of linear flows). Under the same hypotheses as in the theorem, let

$$n_s = \#\{\, \lambda : \lambda \text{ eigenvalue of } DX(x^*),\ \Re(\lambda) < 0 \,\},$$

$$n_u = n - n_s.$$

Then, in a neighborhood of x^*, the flow φ_t is topologically conjugate to the flow of the system

$$\dot{\xi}_s = -\,\xi_s, \qquad \dot{\xi}_u = \xi_u,$$

$$\xi_s \in \mathbb{R}^{n_s},\ \xi_u \in \mathbb{R}^{n_u}.$$

All points in C_0 are equilibria of the fast subsystem. Through the Hartman–Grobman theorem, we may study the normal hyperbolicity and stability of the points in C_0 with respect to the fast variables, which we define as follows:

Definition 2.9. For $m, n \in \mathbb{N}$, given an (m, n)–fast–slow system with critical manifold C_0:

1. The subset $S \subseteq C_0$ is said to be **normally hyperbolic** if, for all $p \in S$,

$$D_x f(p, 0) \text{ has no eigenvalues with zero real part,}$$

where $D_x f$ denotes the Jacobian matrix of f with respect to x. Equivalently, all points $p \in S$ are hyperbolic equilibria of the fast subsystem.

A $p \in S$ is called a normally hyperbolic point of the fast-slow system.

2. A normally hyperbolic set S is said to be **attracting/repelling** if, for all $p \in S$, all eigenvalues of $(D_x f)(p, 0)$ have negative/positive real part, respectively. It is of **saddle** type if $(D_x f)(p, 0)$ has eigenvalues with both positive and negative real parts.

Observation 2.10.
- Note that normal hyperbolicity refers only to the fast directions of the fast subsystem, not full hyperbolicity of the fast-slow system.
- We say that C_0 is normally hyperbolic if all its points are hyperbolic equilibrium points of the fast subsystem.
- In the opposite situation, we say that a point $p \in C_0$ is not normally hyperbolic, or that it is a **loss of normal hyperbolicity**, for the fast–slow system if the matrix $D_x f(p, 0)$ has at least one eigenvalue with zero real part.

We now present Fenichel's theorem, which provides geometric tools and techniques to analyze fast-slow systems with normally hyperbolic submanifolds.

Theorem 2.11. (**Fenichel's theorem**) Suppose that $S_0 \subseteq C_0$ is a compact normally hyperbolic submanifold (possibly with boundary) of the critical manifold C_0. Then, for $\varepsilon > 0$ sufficiently small, the following hold:

(M1) There exists a locally invariant manifold S_ε diffeomorphic to S_0. Local invariance means that trajectories can enter or leave S_ε only through its boundaries.

(M2) S_ε has Hausdorff distance $\mathcal{O}(\varepsilon)$ from S_0 as $\varepsilon \to 0$.

(M3) The flow on S_ε converges to the slow flow as $\varepsilon \to 0$.

(M4) S_ε is C^k-smooth.

(M5) S_ε is normally hyperbolic and has the same stability properties (attracting, repelling, or saddle-type) with respect to the fast variables as S_0.

(M6) S_ε is generally not unique. In regions that remain at a fixed distance from ∂S_ε, all manifolds satisfying *(M1)–(M5)* lie at a Hausdorff distance of order $\mathcal{O}\big(e^{-C/\varepsilon}\big)$ from each other for some constant $C > 0$, with $C = \mathcal{O}(1)$, as $\varepsilon \to 0$.

In qualitative terms, Fenichel's theorem shows that, under its assumptions, a fast-slow system without loss of normal hyperbolicity can be regarded as a regular perturbation of its corresponding singular systems near C_0.

Later on, we will distinguish two cases: First, when the critical manifold C_0 is normally hyperbolic; then, we will be able to apply Fenichel's theorem. The second case appears when we have loss of normal hyperbolicity; then, we will use the blow-up method for a local analysis.

3 BLOW-UP METHOD

This section introduces the general technique of the blow-up method for planar vector fields and illustrates it with a motivating example. We then generalize it to fast–slow systems, which constitute the main object of study in this work.

3.1 Blow-Up of Planar Vector Fields

In the classical literature, the blow-up method is used to desingularize nilpotent equilibrium points of planar vector fields. That is, to study the local dynamics of a system near a non-hyperbolic equilibrium point with null characteristic exponents, by means of a change of variables that transforms the nilpotent equilibrium into hyperbolic ones.

Essentially, the blow-up method is a carefully chosen coordinate transformation in which a degenerate equilibrium is "blown up" into a sphere. In directions transverse to this sphere, and even on the sphere itself, sufficient hyperbolicity is recovered to permit a complete analysis using standard dynamical systems techniques.

We now present the method in six steps and illustrate it with a specific example.

Step 1. Consider a two-dimensional continuous dynamical system:

$$(\dot{x}, \dot{y}) = X(x, y)$$

where $X : U \subseteq \mathbb{R}^2 \to \mathbb{R}^2$ (U open set) is a smooth vector field. Assume that the origin $(0, 0) \in U$ is a nilpotent equilibrium point of X, that is, a non-hyperbolic equilibrium point with all its characteristic exponents equal to zero.

Step 2. Consider a weighted polar change of coordinates

$$\phi : S^1 \times I \to \mathbb{R}^2,$$

where $I \subseteq \mathbb{R}^+$ is an interval containing the origin and S^1 denotes the unit circle.

The change of coordinates ϕ is called the blow-down, while its inverse ϕ^{-1} is called the blow-up. We denote by $\tilde{M} = S^1 \times I$ the cylinder (half-infinite if $I = [0, +\infty)$) and by $B_0 = S^1 \times \{0\}$ the exceptional divisor. The inverse change of coordinates ϕ^{-1} maps the nilpotent equilibrium $(0, 0)$ to the compact set $B_0 \cong S^1$.

Step 3. Apply the change of variables $(x, y) = \phi(\theta, r)$ (blow up), which transforms the planar dynamical system into

$$(\dot{\theta}, \dot{r}) = (D\phi^{-1} \circ X \circ \phi)(\theta, r) \equiv \bar{X}(\theta, r).$$

We assume that the transformed vector field $\bar{X}$, which is called the blow-up of X, can be written as

$$\bar{X}(\theta, r) = \big(r^k\, \Theta(\theta, r),\, r^{k+1} R(\theta, r)\big)$$

for some $k \in \mathbb{N}_0$ and smooth functions $\Theta, R : \tilde{M} \to \mathbb{R}$ such that $\Theta\big|_{B_0}, R\big|_{B_0} \not\equiv 0$.

Note that

$$\bar{X}\big|_{B_0} \equiv (0, 0),$$

that is, the compact set B_0 is entirely filled with equilibria of $\bar{X}$.

Step 4. Apply the reparametrization of time

$$t \mapsto s = s(t), \quad \frac{dt}{ds} = \frac{1}{r^k},$$

which transforms the vector field $\bar{X}$ into

$$(\theta', r') = \frac{1}{r^k}\bar{X}(\theta, r) \equiv \tilde{X}(\theta, r),$$

where $'$ denotes differentiation with respect to s, that is,

$$' = \frac{d}{ds} = \frac{dt}{ds}\frac{d}{dt} = \frac{1}{r^k}\frac{d}{dt}.$$

The new vector field can be written as

$$\tilde{X}(\theta, r) = \big(\Theta(\theta, r),\, rR(\theta, r)\big),$$

and since this is only a time reparametrization, the systems $\bar{X}$ and $\tilde{X}$ have the same phase curves in $\tilde{M} \setminus B_0$.

This reparametrization of time resolves the singularity on B_0, since

$$\tilde{X}\big|_{B_0} \neq (0, 0),$$

i.e., B_0 is no longer entirely filled with equilibria. For this reason, we call this reparametrization the desingularization, and the transformed vector field $\tilde{X}$ the desingularized vector field.

Step 5. Study the stability of the equilibrium points of $\tilde{X}$ on B_0. That is, for those $\theta_0 \in S^1$ such that

$$\Theta(\theta_0, 0) = 0,$$

we analyze the spectrum of the linearization of $\tilde{X}$ at $(\theta, r) = (\theta_0, 0)$:

$$D\tilde{X}(\theta_0, 0) = \begin{pmatrix} \partial_\theta \Theta(\theta_0, 0) & \partial_r \Theta(\theta_0, 0) \\ 0 & R(\theta_0, 0) \end{pmatrix}.$$

We denote

$$\lambda_\theta = \partial_\theta \Theta(\theta_0, 0), \quad \lambda_r = R(\theta_0, 0),$$

the characteristic exponents of the equilibrium in the angular and radial directions, which are tangent (Tg) and transverse (Tr) to B_0, respectively.

If we assume that $(\theta_0, 0)$ is a saddle point (i.e., $\lambda_\theta \lambda_r < 0$), then there exist local stable and unstable invariant manifolds of $\tilde{X}$ associated with $(\theta_0, 0)$ in the tangent and transverse directions to B_0:

$$W^{s,u}_{\tilde{X}}(\theta_0) = W^{Tg,Tr}_{\tilde{X}}(\theta_0).$$

Step 6. Return to the original coordinates by applying the change of variables $(\theta, r) = \phi^{-1}(x, y)$ (blow-down), and use the following result:

Theorem 3.1. (**Planar blow-up theorem**) Assume the previously introduced assumptions and notation. Each local invariant manifold in the transverse direction associated with an equilibrium point $(\theta_0, 0) \in B_0$ of $\tilde{X}$, denoted $W^{Tr}_{\tilde{X}}(\theta_0)$, is mapped by the blow-down map to a local invariant manifold associated with the nilpotent equilibrium point $(0,0) \in U$ of X:

$$W^{\theta_0}_X(0,0) = \phi\big(W^{Tr}_{\tilde{X}}(\theta_0)\big),$$

which preserves the same stability and, at $(0,0)$, is tangent to the direction θ_0.

The application and usefulness of the method will become clearer to the reader through the following example.

Example 3.2. Consider the smooth vector field

$$\begin{aligned}\frac{dx}{dt} &= y,\\ \frac{dy}{dt} &= x^3 + xy,\end{aligned}$$

which has a unique equilibrium point at the origin $(x, y) = (0, 0)$, with Jacobian

$$J = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Since the matrix J has only zero eigenvalues, the origin $(0, 0)$ is a nilpotent non-hyperbolic equilibrium point.

Let us consider a weighted polar change of coordinates

$$\phi : S^1 \times [0, +\infty) \to \mathbb{R}^2, \qquad \phi(\theta, r) = (r\cos\theta, r^2 \sin\theta),$$

which transforms the system into

$$\begin{aligned}\dot{\theta} &= \frac{r\left(1 + \sin\theta - 4\sin^2\theta - \sin^3\theta + \sin^4\theta\right)}{\sin^2\theta + 1},\\ \dot{r} &= \frac{r^2(\cos\theta\sin\theta)(\sin\theta - \sin^2\theta + 2)}{\sin^2\theta + 1}.\end{aligned}$$

The desingularized system is therefore

$$\begin{aligned}\theta' &= \frac{1 + \sin\theta - 4\sin^2\theta - \sin^3\theta + \sin^4\theta}{\sin^2\theta + 1},\\ r' &= \frac{r(\cos\theta\sin\theta)(\sin\theta - \sin^2\theta + 2)}{\sin^2\theta + 1}.\end{aligned}$$

This system has four saddle (hyperbolic) equilibrium points on B_0:

$$\begin{aligned} p_1 &= \Big(-\arcsin(\sqrt{2} - 1),\, 0\Big), & p_2 &= \Big(\arcsin(\sqrt{5/2} - 1/2),\, 0\Big),\\ p_3 &= \Big(\pi - \arcsin(\sqrt{5/2} - 1/2),\, 0\Big), & p_4 &= \Big(\pi + \arcsin(\sqrt{2} - 1),\, 0\Big).\end{aligned}$$

The dynamics of the desingularized system around the saddles on B_0 is illustrated in Figure 1. In the figure, we represent the half-infinite cylinder $\tilde{M} = S^1 \times [0, +\infty)$ via the homeomorphism

$$\varphi : S^1 \times \mathbb{R} \to \mathbb{R}^2 \setminus \{(0,0)\}, \quad \varphi(\theta, r) = (e^r \cos\theta, e^r \sin\theta),$$

which maps

$$\varphi(\tilde{M}) = \mathbb{R}^2 \setminus \mathring{\mathbb{D}}^2 \quad \text{and} \quad \varphi(B_0) = S^1,$$

providing a planar representation of the cylinder.

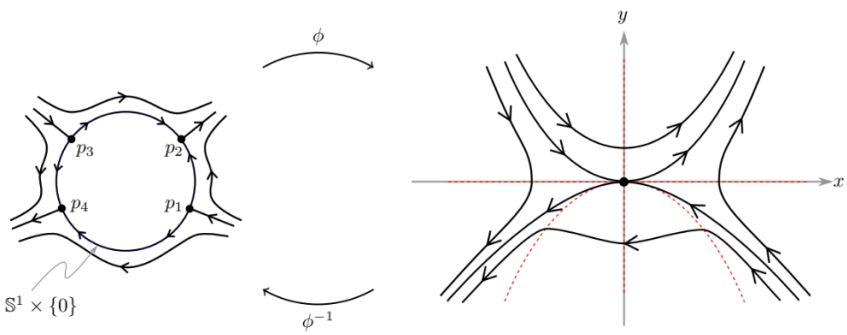

Figure 1: On the left, we show the planar representation of the phase portrait of the desingularized system for the considered example, around the four saddles $p_i \in B_0$. On the right, we show the phase portrait of the original system around the nilpotent equilibrium $(0,0)$. The arrows indicate the blow-down ϕ and blow-up ϕ^{-1} associated with this process. The phase portrait on the right is obtained from that on the left via the blow-up method.

3.2 Blow-Up of Fast–Slow Systems

In the case of fast–slow systems, as defined in (2.2) (fast time-scale), the method is similar, but needs to be generalized to a higher dimension.

Let us first rewrite the ε-family of vector fields on $\mathbb{R}^{m+n}$ as a single vector field on $\mathbb{R}^{m+n+1}$, given by

$$\begin{aligned} \dot{x} &= f(x, y, \varepsilon), \\ \dot{y} &= \varepsilon\, g(x, y, \varepsilon), \\ \dot{\varepsilon} &= 0. \end{aligned} \tag{3.1}$$

Following the standard case that we will be studying at this work, we can assume that the origin $(x, y, \varepsilon) = (0, 0, 0)$ is an equilibrium point of (3.1), and that $D_x f(0, 0, 0)$ has all its eigenvalues equal to zero. Thus, the origin is a nilpotent singularity of the system, and the blow-up method can be applied to desingularize it.

We now present the quasihomogeneous blow-up, which is quite commonly used in the literature. We generalize the concepts introduced in the planar case to the fast–slow system framework, but the method remains the same.

Consider the smooth vector field

$$X : \mathbb{R}^{m+n+1} \to \mathbb{R}^{m+n+1}$$

defined by (3.1), and assume that $X(0, 0, 0) = (0, 0, 0)$ is a nilpotent equilibrium point.

Definition 3.3. We define the **generalized weighted polar transformation** as the map

$$\Phi : S^{m+n} \times I \to \mathbb{R}^{m+n+1}, \quad \Phi(\omega, r) = \Phi(\bar{x}, \bar{y}, \bar{\varepsilon}, r) = \big(r^{\alpha}\bar{x},\, r^{\beta}\bar{y},\, r^{\gamma}\bar{\varepsilon}\big), \tag{3.2}$$

where S^{m+n} is the unit sphere in $\mathbb{R}^{m+n+1}$ and $I \subseteq \mathbb{R}^{+}$ is an interval containing zero. Moreover,

$$\omega = (\bar{x}, \bar{y}, \bar{\varepsilon}) = (\bar{x}_1, \ldots, \bar{x}_m, \bar{y}_1, \ldots, \bar{y}_n, \bar{\varepsilon}) \in S^{m+n}, \quad r \in I,$$

and we use the multi-index notation

$$r^{\alpha}\bar{x} = (r^{\alpha_1}\bar{x}_1, \ldots, r^{\alpha_m}\bar{x}_m),$$

for $\alpha = (\alpha_1, \ldots, \alpha_m) \in \mathbb{N}_0^m$, $\beta = (\beta_1, \ldots, \beta_n) \in \mathbb{N}_0^n$, and $\gamma \in \mathbb{N}_0$.

The change of coordinates Φ is called the **blow-down**, while its inverse Φ^{-1} is called the **blow-up**. We denote by $\tilde{M} = S^{m+n} \times I$ the generalized cylinder (half-infinite if $I = [0, +\infty)$) and by $B_0 = S^{m+n} \times \{0\}$ the exceptional divisor. The inverse change of coordinates Φ^{-1} maps the nilpotent equilibrium $(0, 0, 0)$ to the compact sphere $B_0 \cong S^{m+n}$.

Definition 3.4. The **quasihomogeneous blow-up vector field** of X, denoted by $\bar{X}$, is defined as

$$\bar{X}(\omega, r) = (D\Phi^{-1} \circ X \circ \Phi)(\omega, r) = \big(r^k\, \Omega(\omega, r),\, r^{k+1}R(\omega, r)\big), \tag{3.3}$$

for some $k \in \mathbb{N}_0$ and smooth functions

$$\Omega : \tilde{M} \to TS^{m+n}, \qquad R : \tilde{M} \to \mathbb{R},$$

where TS^{m+n} denotes the tangent bundle of the sphere S^{m+n}, and such that $\Omega\big|_{B_0} \not\equiv 0$ and $R\big|_{B_0} \not\equiv 0$.

Observation 3.5. "Quasihomogeneous" refers to the fact that after the blow-up, the vector field components scale with different powers of the radial variable, unlike fully homogeneous vector fields where all components scale with the same power.

Definition 3.6. The **desingularized vector field** of X, denoted by $\tilde{X}$, is defined as

$$\tilde{X}(\omega, r) = \frac{1}{r^k}\,\bar{X}(\omega, r) = \big(\Omega(\omega, r),\, rR(\omega, r)\big),$$

obtained by applying the time reparametrization

$$t \mapsto s = s(t), \quad \frac{dt}{ds} = \frac{1}{r^k},$$

to the blow-up vector field $\bar{X}$.

Observation 3.7. The dynamical systems defined by $\bar{X}$ and $\tilde{X}$ are conjugate and equivalent to X on $\tilde{M} \setminus B_0$, respectively. For the dynamics on the exceptional divisor $B_0 \cong S^{m+n}$, note that B_0 is entirely filled with equilibrium points of $\bar{X}$, while only those directions ω satisfying $\Omega(\omega, 0) = 0$ are equilibrium points of $\tilde{X}$ (desingularization).

Typically, $\tilde{X}$ has semi-hyperbolic singularities, hyperbolic singularities, or no singularities at all. Therefore, due to the equivalences between X, $\bar{X}$, and $\tilde{X}$, we conclude that for $r > 0$ sufficiently small, the vector field $\tilde{X}$ provides a complete description of the phase portrait of X around the nilpotent equilibrium.

Formally, and making use of the center manifold theorem, we have:

Theorem 3.8 (**Generalized blow-up theorem for fast–slow systems**)**.** Assume the previously introduced assumptions and notation. Let $(\omega_0, 0) \in B_0 = S^{m+n} \times \{0\}$ be an equilibrium of $\tilde{X}$, with transverse characteristic exponent

$$\lambda_r = R(\omega_0, 0) \neq 0.$$

Let $W^{Tr}_{\tilde{X}}(\omega_0) \subseteq \tilde{M}$ denote the associated one-dimensional local invariant manifold of $\tilde{X}$, which is stable if $\lambda_r < 0$ and unstable if $\lambda_r > 0$.

Then, under the blow-down map Φ, this invariant manifold is mapped to a local invariant manifold of the original system X associated with the nilpotent equilibrium $(0, 0, 0) \in \mathbb{R}^{m+n+1}$:

$$W^{\omega_0}_X(0, 0, 0) = \Phi\big(W^{Tr}_{\tilde{X}}(\omega_0)\big),$$

preserving the same stability type. Moreover, at $(0, 0, 0)$, the tangent space of $W^{\omega_0}_X(0, 0, 0)$ coincides with the direction ω_0 on the sphere S^{m+n}.

To conclude this section, we briefly comment on the case of higher-dimensional problems, namely when $m + n > 2$. In such situations, working with polar coordinates quickly becomes cumbersome. It is therefore convenient to work with charts that cover the blow-up space. Within each chart, one introduces local coordinates together with a corresponding local vector field.
In practice, local coordinates are defined by fixing one of the spherical blow-up coordinates equal to ± 1. This procedure is commonly referred to as a directional blow-up.

Definition 3.9. Let $(\bar{x}_0, \bar{y}_0, \bar{\varepsilon}_0) \in S^{m+n}$ be a fixed direction on the sphere. Recalling the quasihomogeneous blow-up map (3.2), we define

$$\Phi_{(\bar{x}_0, \bar{y}_0, \bar{\varepsilon}_0)}(r) := \Phi(\bar{x}_0, \bar{y}_0, \bar{\varepsilon}_0, r).$$

This map is called the **directional blow-up** along the direction $(\bar{x}_0, \bar{y}_0, \bar{\varepsilon}_0)$.

Example 3.10. For instance, a blow-up in the ε-direction is given by $\Phi(0, 0, 1, r)$ or $\Phi(0, 0, -1, r)$. For a two-dimensional fast–slow system, a blow-up in the fast direction corresponds to $\Phi(1, 0, 0, r)$ or $\Phi(-1, 0, 0, r)$, while a blow-up in the slow direction corresponds to $\Phi(0, 1, 0, r)$ or $\Phi(0, -1, 0, r)$.

Directional blow-ups give rise to local vector fields on the corresponding Euclidean charts. After analyzing the dynamics of these local vector fields, one can identify overlapping regions between charts and relate the corresponding flows through so-called matching (or transition) maps. This procedure allows for a global description of the dynamics on $S^{m+n} \times I$. In particular, it enables the continuation and tracking of invariant objects—most notably center manifolds—throughout the blow-up space. A schematic illustration of the blow-up construction is shown in Figure 2.

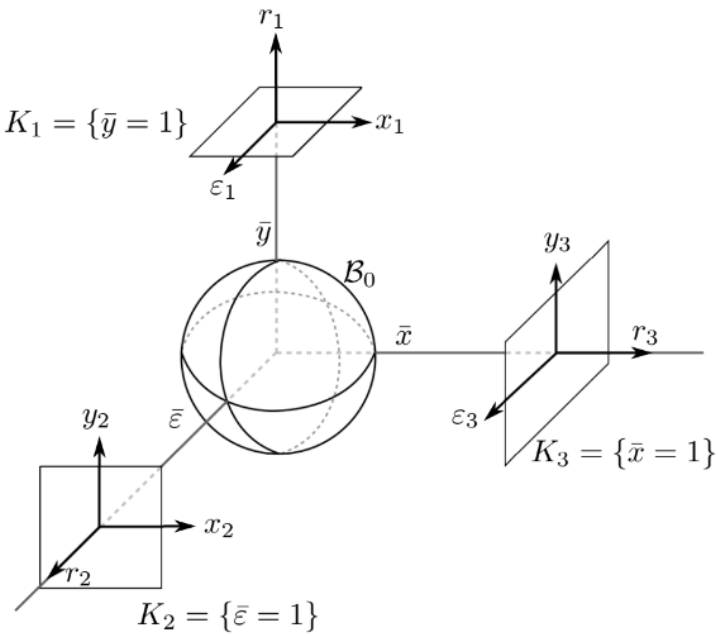

Figure 2: Schematic representation of the blow-up construction and its directional charts. Three distinguished directions are shown, corresponding to the fast direction $\bar{x}$, the slow direction $\bar{y}$, and the ε-direction $\bar{\varepsilon}$. The sphere B_0 represents the exceptional divisor.

4 RELAXATION OSCILLATIONS

In this section, we introduce the phenomenon of relaxation oscillations, which are a class of periodic solutions arising in singularly perturbed systems.

Relaxation oscillations are oscillatory motions characterized by the alternation of two distinct time scales: long intervals of slow, quasi-static evolution during which the system follows attracting branches of a reduced (slow) dynamics, separated by short intervals of rapid transition in which the solution undergoes fast jumps between these branches. Although relaxation oscillations can be found in a more general situation, we will study them in the following context:

Definition 4.1. Consider an (m, n)-fast–slow system written in the slow time scale τ:

$$\begin{cases} \varepsilon \frac{dx}{d\tau} = f(x, y, \varepsilon), \\ \frac{dy}{d\tau} = g(x, y, \varepsilon), \end{cases} \qquad x \in \mathbb{R}^m,\ y \in \mathbb{R}^n,\ 0 < \varepsilon \ll 1, \tag{4.1}$$

with critical manifold:

$$S_0 = \{(x, y) \in \mathbb{R}^m \times \mathbb{R}^n :\ f(x, y, 0) = 0\}.$$

A **relaxation oscillation** is a family of T_ε-periodic solutions $(x_\varepsilon(\tau), y_\varepsilon(\tau))$ such that, as $\varepsilon \to 0$:

i) On most of the period, the trajectory remains $O(\varepsilon)$-close to normally hyperbolic attracting or repelling branches of S_0 and evolves on the slow time scale according to the reduced equations

$$0 = f(x, y, 0), \qquad \frac{dy}{d\tau} = g(x, y, 0), \quad (x, y) \in S_0.$$

ii) At finitely many jump times, the solution rapidly transitions between branches of S_0. In the fast time scale $t = \tau/\varepsilon$, these transitions are described by the layer equations

$$\frac{dx}{dt} = f(x, y, 0), \qquad \frac{dy}{dt} = 0,$$

with y frozen.

iii) The concatenation of slow reduced-flow segments on S_0 and fast layer-flow jumps forms a closed curve in phase space.

Observation 4.2.

- A more precise statement, complementary to the above definition, is that the family of trajectories $(x_\varepsilon, y_\varepsilon)$ converges (in the Hausdorff sense) to a singular limit cycle consisting of slow arcs on S_0 and instantaneous jumps.
- Here, we have described the qualitative behavior of relaxation oscillations in fast–slow systems. A next step is to prove that such families of periodic solutions actually exist.

A natural question is how these fast jumps between segments of S_0 arise. The answer is that these jumps occur at fold points of the fast–slow system. We now define them in this context:

Definition 4.3. A point $p \in S_0$ is called a **fold point** of the fast–slow system if:

1. Loss of normal hyperbolicity:
$$\operatorname{rank} D_x f(p, 0) = m - 1,$$
i.e., p has zero as a simple characteristic exponent for the fast subsystem.

2. Nondegeneracy of the fold in the fast direction: Let $v \in \ker D_x f(p, 0)$ be the eigenvector of eigenvalue zero. Then
$$v^\top D_x^2 f(p, 0)[v, v] \neq 0,$$
or in Einstein summation notation,
$$\frac{\partial^2 f_i}{\partial x_j \partial x_k}(p, 0) v_i \, v_j \, v_k \neq 0.$$

Observation 4.4. We observe that fold points are equilibria of the fast subsystem where normal hyperbolicity is lost in exactly one fast direction and the critical manifold locally bends. Further regularity conditions can be imposed on fold points:

Definition 4.5. A fold point p is said to be **generic** (or regular) if also satisfies a transversality condition with respect to the slow flow:

$$D_y f(p, 0)\, g(p, 0) \notin \operatorname{Range}(D_x f(p, 0)),$$

i.e., the slow flow is not tangent to the fast directions at the fold.

Observation 4.6. The intuition behind this definition is that a generic fold ensures that trajectories of the full system actually reach the fold and are "pushed off" along a fast jump. Non-generic folds may have the slow flow tangent to the fast direction at the fold or higher-order degeneracies, leading to canard phenomena or more complicated dynamics.

Example 4.7. For the FitzHugh-Nagumo model (2.6), one can observe relaxation oscillations for a ε sufficient small. In fact, we can approximate the period of the relaxation oscillations according to the expression:

$$T = \frac{1}{\varepsilon}\int_{W_-}^{W_+}\left(\frac{1}{G_+(w)-\gamma w} - \frac{1}{G_-(w)-\gamma w}\right)dw,$$

where $G_-(w)$ and $G_+(w)$ are the expressions of the two slow branches of the cubic- shaped nullcline $\dot{v}=0$, and W_- (resp., W_+) is the local minimum (resp., maximum) of this cubic.

From now on, we **focus on** $(1,1)$**-fast–slow systems**, and we try to characterize the simplest type of relaxation oscillations. We assume that the system has two folds, which satisfy the condition $f_x(x_0,y_0,0)=0$, that is, they are critical points of the critical manifold S_0, which we parametrize by the curve $y=\varphi(x)$. We assume these critical points are non-degenerate, and constitute a minimum at the origin, and a maximum at $x=x_M>0$. Thus, we can write the critical manifold S_0 as the union of:

$$\begin{aligned} S_l &= \{(x,\varphi(x)) : x<0\}, \\ S_m &= \{(x,\varphi(x)) : 0<x<x_M\}, \\ S_r &= \{(x,\varphi(x)) : x>x_M\}. \end{aligned} \tag{4.2}$$

Moreover, we assume that:

1. For the layer problem, S_l and S_r are attracting, i.e. $f_x<0$ on S_l and S_r, and S_m is repelling, i.e. $f_x>0$ on S_m.

2. Both folds are generic, i.e., satisfy the following conditions:

$$f_{xx}(x_0,y_0,0)\neq 0, \quad f_y(x_0,y_0,0)\neq 0, \quad g(x_0,y_0,0)\neq 0.$$

3. Substituting $y=\varphi(x)$ into the reduced equation (2.3), we obtain

$$\dot{x} = \frac{g(x,\varphi(x),0)}{\varphi'(x)}\,. \tag{4.3}$$

   Then, the slow flow on S_l satisfies $\dot{x}>0$, and the slow flow on S_r satisfies $\dot{x}<0$.

We end having the fast–slow dynamics represented in Figure 3.

4.1 Existence of Relaxation Oscillations

Now we aim to present an important theorem concerning the existence of a limit cycle in the context of relaxation oscillations. To this end, let us denote by U a neighborhood of the origin. Note that our S-shaped critical manifold, restricted to this neighborhood, has two branches: $S_{U_a}\subset S_l$ and $S_{U_r}\subset S_m$, corresponding respectively to the attracting and repelling parts.

We observe that the point $(0,0)$ is the only point in $U\cap S_0$ at which

$$\frac{\partial f}{\partial x}=0,$$

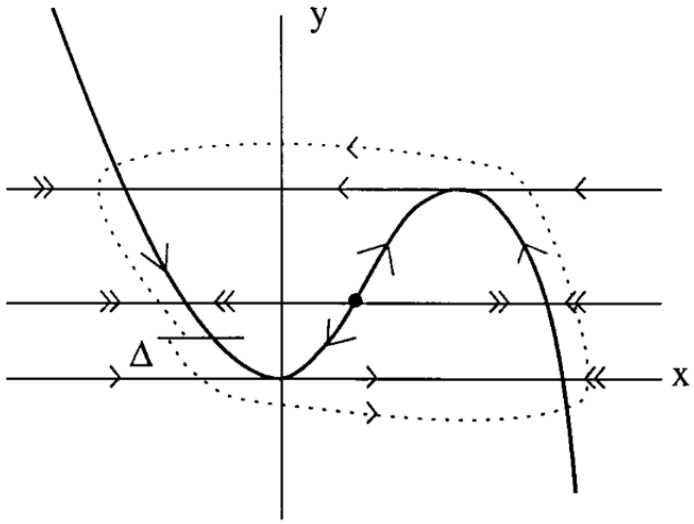

Figure 3: The S-shaped critical manifold (solid line) and the relaxation oscillation orbit (dotted line) for the considered (1,1)-fast–slow system. The arrows along the critical manifold indicate the slow flow direction, and the black dot a (unstable) slow equilibrium of the system.

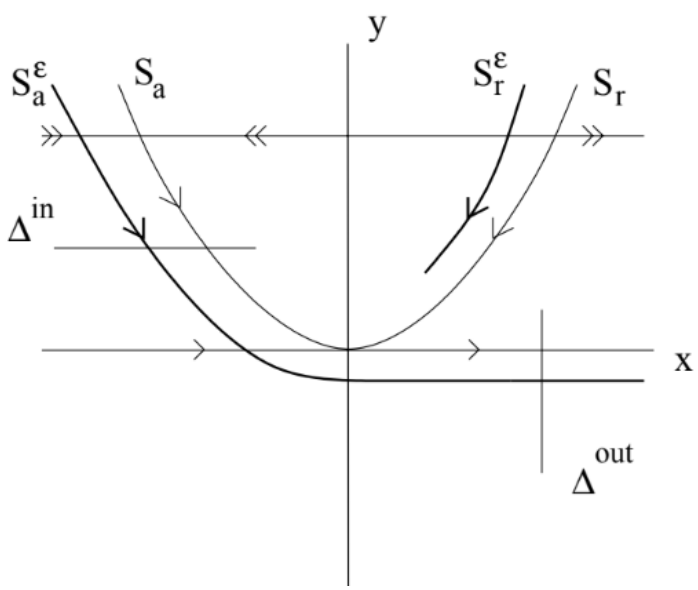

Figure 4: Critical manifold, slow manifolds and sections for the fold point.

and that $U \cap S_0$ is approximately parabolic. Let S_a (resp. S_r) denote its left (resp. right) branch, so that

$$S_0 \cap U = S_{U_a} \cup S_{U_r}$$

(see Figure 4).

We may assume without loss of generality that, for the layer problem, the fold at $(0,0)$ satisfies

$$\frac{\partial^2 f}{\partial x^2}(0,0,0) > 0, \qquad \frac{\partial f}{\partial y}(0,0,0) < 0.$$

The origin is non-hyperbolic, weakly attracting from the left and weakly repelling to the right. The reduced dynamics is then determined by (4.3) and is singular at $x = 0$. Our assumptions on f and g imply that the direction of the reduced flow is determined by the sign of $g(0,0,0)$. We assume that

$$g(0,0,0) < 0.$$

This implies that the reduced flow on S_{U_a} and S_{U_r} is directed towards the fold point, as shown again in Figure 4. In fact, orbits on S_{U_a} and S_{U_r} reach the fold point in finite time due to the singularity of the reduced flow at the fold point.

The only possibility to continue from there in the singular limit is along the (weakly) unstable fiber of the layer problem along the positive x-axis. Thus, the curve

$$S_{U_a} \cup \{(x,0) : x > 0\}$$

is expected to be a zeroth-order approximation. This is the situation relevant to relaxation oscillations; we refer to this case as a **jump point**.

The case $g(0,0,0) > 0$ can be analyzed similarly. From standard Fenichel theory [1] we know that, outside an arbitrarily small neighborhood V of $(0,0)$, the manifolds S_{U_a} and S_{U_r} perturb smoothly to locally invariant manifolds $S_{U_a,\varepsilon}$ and $S_{U_r,\varepsilon}$ for sufficiently small $\varepsilon > 0$. Notice that $S_{U_a,\varepsilon}$ and $S_{U_r,\varepsilon}$ are actually very simple, as they consist of single solutions. Moreover, the slow manifolds are obtained as sections $\varepsilon =$ const. of two-dimensional, locally invariant, center-like manifolds M_a (resp. M_r) of the extended system

$$\begin{cases} \dot{x} = f(x,y,\varepsilon), \\ \dot{y} = \varepsilon g(x,y,\varepsilon), \\ \dot{\varepsilon} = 0. \end{cases} \tag{4.4}$$

In the extended phase space $\mathbb{R}^3$, the set $S_0 \times \{0\}$ is a manifold of equilibria for the extended system. Outside a neighborhood of the fold point $(0,0,0)$, the linearization of system (4.4) at points in $S_a \times \{0\}$ has a double zero eigenvalue and one uniformly hyperbolic (stable) eigenvalue. This allows us to conclude the existence of the attracting center-like manifold M_a; the manifold M_r is obtained similarly. At the fold point $(0,0,0)$, the linearization has a triple zero eigenvalue, and the construction of the slow manifolds breaks down.

We focus our attention on S_a and investigate how $S_{a,\varepsilon}$, as well as nearby solutions, behave as they pass near the fold point. We expect that, close to the fold point, a transition from slow motion along $S_{a,\varepsilon}$ to fast motion almost parallel to the unstable fibers occurs. A similar analysis could be carried out for $S_{r,\varepsilon}$.

Observation 4.8. It is known that the slow manifolds M_a and M_r, and hence their sections $S_{a,\varepsilon}$ and $S_{r,\varepsilon}$, are not unique and are determined only up to $\mathcal{O}(e^{-k/\varepsilon})$, where k is some positive constant. We make an arbitrary choice of M_a and M_r and indicate at the end that our results are independent of this choice.

We now view the previously introduced neighborhood U as a neighborhood of $(0,0,0)$ in $\mathbb{R}^3$. We choose U sufficiently small so that $g(x,y,\varepsilon)=0$ for $(x,y,\varepsilon)\in U$.

Before stating the main results, we rewrite system (4.1) (respectively, (4.4)) in canonical form. By rescaling x, y, ε, and t, we obtain

$$\begin{aligned}\dot{x} &= -y + x^2 + h(x,y,\varepsilon),\\ \dot{y} &= \varepsilon g(x,y,\varepsilon),\\ \dot{\varepsilon} &= 0,\end{aligned} \tag{4.5}$$

where

$$h(x,y,\varepsilon) = \mathcal{O}(\varepsilon, xy, y^2, x^3), \qquad g(x,y,\varepsilon) = -1 + \mathcal{O}(x,y,\varepsilon),$$

and the new function g is related to the original one by the rescaling. This form of the equations will be used throughout the forthcoming analysis.

For small $\rho > 0$ and a suitable interval $J \subset \mathbb{R}$, let

$$\Delta_{\text{in}} = \{(x,\rho^2) \mid x \in J\}$$

be a section in U transverse to S_a, and let

$$\Delta_{\text{out}} = \{(\rho, y) \mid y \in \mathbb{R}\}$$

be a section in U transverse to the fast fibers. Let

$$\pi\colon \Delta_{\text{in}} \to \Delta_{\text{out}}$$

denote the transition map induced by the flow of the slow–fast system.

Theorem 4.9 (Existence of the limit cycle). Under the assumptions made in this section, there exists $\varepsilon_0 > 0$ such that the following assertions hold for all $\varepsilon \in (0, \varepsilon_0]$:

1. The manifold $S_{a,\varepsilon}$ passes through Δ_{out} at a point $(\rho, h(\varepsilon))$, where $h(\varepsilon) = \mathcal{O}(\varepsilon^{2/3})$.
2. The transition map π is a contraction with contraction rate $\mathcal{O}(e^{-k/\varepsilon})$, where k is a positive constant.

Proof. For a detailed proof, we refer the reader to [7]. □

Let $(x_r, 0)$ denote the point where the branch S_r of the critical manifold intersects the x-axis, and let (x_ℓ, y_M) denote the point where the branch S_ℓ intersects the horizontal line $y = y_M$.

We define the **singular trajectory** Γ_0 as the union of:

- the critical fibers connecting $(0,0)$ to $(x_r, 0)$ and (x_M, y_M) to (x_ℓ, y_M),

- the two segments of the critical manifold connecting $(x_r, 0)$ to (x_M, y_M) and (x_ℓ, y_M) to $(0, 0)$.

Let U be a small tubular neighborhood of Γ_0. A continuous family of periodic orbits Γ_ε is called a **family of relaxation oscillations** if

$$\Gamma_\varepsilon \longrightarrow \Gamma_0 \quad \text{in the Hausdorff distance as } \varepsilon \to 0.$$

The following well-known result establishes the existence of such a family of relaxation oscillators.

Theorem 4.10 (**Existence of a family of relaxation oscillators**)**.** Assume the hypotheses of this section. Then, for sufficiently small $\varepsilon > 0$, there exists a unique limit cycle $\Gamma_\varepsilon \subset U$. The cycle Γ_ε is strongly attracting; more precisely, its Floquet exponent is bounded above by $-K/\varepsilon$, where $K > 0$ is a constant. Moreover,

$$\Gamma_\varepsilon \longrightarrow \Gamma_0 \quad \text{in the Hausdorff distance as } \varepsilon \to 0.$$

Proof. By Fenichel theory, the branches S_ℓ and S_r perturb to slow manifolds $S_{\ell,\varepsilon}$ and $S_{r,\varepsilon}$. By Theorem 4.9, the manifolds $S_{\ell,\varepsilon}$ and $S_{r,\varepsilon}$ continue beyond their respective fold points and follow, to leading order, the critical fibers, arriving in the vicinity of S_r and S_ℓ, respectively.

Let Σ be a section of the flow defined as a small horizontal interval intersecting S_ℓ at a point between (x_ℓ, y_M) and $(0, 0)$. Consider a trajectory starting in Σ for $0 < \varepsilon \ll 1$. Initially, this trajectory is attracted to $S_{\ell,\varepsilon}$ and then, by Theorem 4.9, to the extension of $S_{\ell,\varepsilon}$ beyond the fold point at $(0, 0)$. When it arrives in the vicinity of S_r, it is attracted to $S_{r,\varepsilon}$ and follows it and its extension until it approaches S_ℓ. It then follows $S_{\ell,\varepsilon}$ until it returns to Σ.

Let $P : \Sigma \to \Sigma$ denote the resulting return map. It follows from Theorem 4.9 and Fenichel theory that, for ε sufficiently small, P is a contraction with contraction rate bounded above by $e^{-K/\varepsilon}$ for some constant $K > 0$. By the implicit function theorem, P possesses a unique, strongly attracting fixed point in Σ. This fixed point corresponds to a limit cycle Γ_ε. Moreover, the Floquet exponent of Γ_ε is bounded above by $-K/\varepsilon$.

Let $q_\varepsilon \in \Sigma \cap S_{\ell,\varepsilon}$, and let Γ'_ε denote the portion of the forward trajectory of q_ε up to its first return to Σ. By Theorem 4.9, the curve Γ'_ε approaches the singular cycle Γ_0 as $\varepsilon \to 0$. Since Γ'_ε is exponentially close to Γ_ε, it follows that Γ_ε approaches Γ_0 in the Hausdorff distance as $\varepsilon \to 0$. $\square$

5 CANARDS

In this section, we define canard trajectories in the context of planar fast–slow systems and introduce the phenomenon of canard explosion. This type of dynamics arises in classical systems such as the Van der Pol oscillator and the FitzHugh–Nagumo model.

Let us start by defining the concept of a canard:

Definition 5.1. A **canard point** is a non-generic fold point of a fast–slow system. In the case of a $(1,1)$ fast–slow system, a canard point (x_0, y_0) satisfies

$$f(x_0, y_0, 0) = 0, \quad f_x(x_0, y_0, 0) = 0, \quad f_{xx}(x_0, y_0, 0) \neq 0, \quad \text{and} \quad g(x_0, y_0, 0) = 0.$$

Observation 5.2.

- A canard point is a fold at which the slow flow is tangent to the critical manifold in the degenerate fast direction. At such points, the generic transversality condition fails, and attracting and repelling slow manifolds meet.
- In contrast to generic folds, the fast–slow system does not jump immediately between branches of the critical manifold near a canard point. Instead, trajectories may follow a repelling slow manifold for a $\mathcal{O}_\tau(1)$ time. Such trajectories are called **canards**.
  Near the canard point, the attracting and repelling slow manifolds $S_{a,\varepsilon}$ and $S_{r,\varepsilon}$, intersect for a sufficiently small ε. The intersection creates the canard trajectories near the point. That trajectory that follows the repelling manifold the longest in the maximal canard, and the rest of nearby solutions are secondary canards.

If we assume, in the planar canard setting, that the system has a fold point (x_0, y_0), then for sufficiently small $\varepsilon > 0$, the full fast–slow system has an equilibrium $(x_\varepsilon, y_\varepsilon)$ close to the fold point, which can be a focus, a node or a saddle in general.

The equilibria $(x_\varepsilon, y_\varepsilon)$ is typically off the critical manifold by a $\mathcal{O}(\varepsilon)$ amount. As $\varepsilon \to 0$, we have that $(x_\varepsilon, y_\varepsilon) \to (x_0, y_0)$.

In particular, if the fold is a canard point, then the dynamics of the system close to the equilibria exhibit canard trajectories as $\epsilon \to 0$.

5.1 Canard Explosion

Now, consider the parametric family of $(1,1)$ fast–slow systems depending on a parameter $\lambda \in \mathbb{R}$:

$$\begin{aligned} \dot{x} &= f(x, y, \lambda, \varepsilon), \\ \dot{y} &= \varepsilon g(x, y, \lambda, \varepsilon), \end{aligned} \tag{5.1}$$

where we assume f and g to be smooth with respect to all parameters and variables (at least C^3).

Such a parametrized family may exhibit a dynamical phenomenon known as a **canard explosion**. Qualitatively, it is a transition in the dynamics of the fast–slow system as λ changes: small-amplitude periodic orbits suddenly increase their amplitude as the parameter λ approaches some threshold value λ_c, giving rise to relaxation oscillations. Let us explain under which assumptions such phenomena can occur:

- Let us assume that, for $\lambda = 0$, the fast–slow system satisfies the conditions we introduced in Section 4. In particular, the critical manifold S_0 has a minimum at $(0,0)$ and a maximum at

(x_M, y_M) (fold points), and decomposes into the subsets (4.2). We also assume that S_l and S_r are attracting, i.e. $f_x < 0$ on S_l and S_r, while S_m is repelling, i.e. $f_x > 0$ on S_m.

- Assume that, for each λ, the fast–slow system has an equilibrium point $q_{\lambda,\varepsilon}$ close to a fold point p_λ. We recall that $q_{\lambda,\varepsilon}$ is off S_0 by an $\mathcal{O}(\varepsilon)$ amount in the fast direction, and collapses onto p_λ as $\varepsilon \to 0$.

- Assume that the equilibrium point $q_{\lambda,\varepsilon}$ undergoes a Hopf bifurcation for $\lambda = \lambda_H(\varepsilon)$. We recall that, in a Hopf bifurcation, a stable focus transitions into an unstable focus as the parameter crosses a threshold value, and a stable limit cycle emerges, with amplitude proportional to $\sqrt{|\lambda - \lambda_H(\varepsilon)|}$ for λ close to $\lambda_H(\varepsilon)$.
  This type of Hopf bifurcation is called a **singular Hopf**, since it occurs near the fold of the critical manifold.

- Assume that the fold point p_λ becomes a canard point for $\lambda = \lambda_c$. Without loss of generality, we assume $\lambda_c = 0$. This means that, for this value of λ, the fold loses genericity, and canard trajectories may emerge as $\varepsilon \to 0$ for this parameter value. As $\varepsilon \to 0$, we have $\lambda_H(\varepsilon) \to \lambda_c = 0$. More precisely,
  $$\lambda_H(\varepsilon) = \lambda_c + \mathcal{O}\big(e^{-k/\varepsilon}\big), \quad k > 0.$$
  For instance, we can take as canard point the origin $p_0 = (0,0)$. We also assume some non-degeneracy conditions for the canard point: at $(x, y, \lambda, \varepsilon) = (0,0,0,0)$, in total, we are assuming
  $$f(0,0,0,0) = 0, \quad g(0,0,0,0) = 0, \quad f_x(0,0,0,0) = 0, \tag{5.2}$$
  $$f_{xx}(0,0,0,0) \neq 0, \quad f_y(0,0,0,0) \neq 0, \quad g_x(0,0,0,0) \neq 0, \quad g_\lambda(0,0,0,0) \neq 0. \tag{5.3}$$

Under these assumptions, we can study how the dynamics of the system changes as λ varies:

- For the values of λ for which the focus equilibrium point $q_{\lambda,\varepsilon}$ is unstable, for instance $\lambda \gg 0$, we observe small-amplitude oscillations which are repelled by the equilibrium and attracted by the stable limit cycle.

- As $\varepsilon \to 0$ and $\lambda \to \lambda_c = 0$, since the fold approximates to a canard point, these small-amplitude periodic orbits partly follow the repelling branch after crossing the fold, and delay their jump to the stable branches, producing large-amplitude oscillations known as **canard cycles** (see Figure 5).

The amplitude of the canard cycles increases very rapidly to $\mathcal{O}(1)$ when λ enters an exponentially small window around $\lambda_c = 0$:
$$|\lambda - \lambda_c| = \lambda = \mathcal{O}\big(e^{-k/\varepsilon}\big).$$
This is why this phenomenon is interpreted as an "explosion," which leads to relaxation oscillations in a fast–slow system.

In the considered context, let us now introduce a parametrization for the family of closed singular trajectories that generate the canard cycles.

First, concerning the flow on $S_l \cup \{0\} \cup S_m$ and the flow on S_r, we will assume that, when $\lambda = 0$, we have $\dot{x} < 0$ for the slow flow on S_r and $\dot{x} > 0$ for the slow flow on $S_l \cup \{0\} \cup S_m$. This situation is precisely the one for which, by Theorem (4.10), a relaxation oscillation exists for $\varepsilon > 0$.

Let $y = \varphi(x)$ be the parametrization of the critical manifold, and set $\lambda = 0$. Let $y_M = \varphi(x_M)$. For $s \in (0, y_M)$, let

$$x_\ell(s) < x_m(s) < x_r(s)$$

be the three distinct roots of $\varphi(x) = s$. We set

$$x_\ell(0) = x_m(0) = 0 \quad \text{and} \quad x_m(y_M) = x_r(y_M) = x_M.$$

We then parametrize canard cycles as follows: For $s \in [0, y_M]$, define

$$\Gamma(s) = \big\{(x, \varphi(x)) : x \in [x_\ell(s), x_m(s)]\big\} \cup \big\{(x, s) : x \in [x_\ell(s), x_m(s)]\big\}.$$

For $s \in [y_M, 2y_M]$, define

$$\begin{aligned}\Gamma(s) = &\big\{(x, \varphi(x)) : x \in [x_\ell(y_M), x_m(2y_M - s)]\big\} \cup \big\{(x, 2y_M - s) : x \in [x_m(2y_M - s), x_r(2y_M - s)]\big\} \\ &\cup \big\{(x, \varphi(x)) : x \in [x_M, x_r(2y_M - s)]\big\} \cup \big\{(x, y_M) : x \in [x_\ell(y_M), x_M]\big\}.\end{aligned}$$

We illustrate two representative members of the family of singular cycles $\Gamma(s)$ in Figure 5. Cycles in this family with $s \in (0, y_M)$ and $s \in (y_M, 2y_M)$ are commonly referred to as **canards without head** and **canards with head**, respectively.

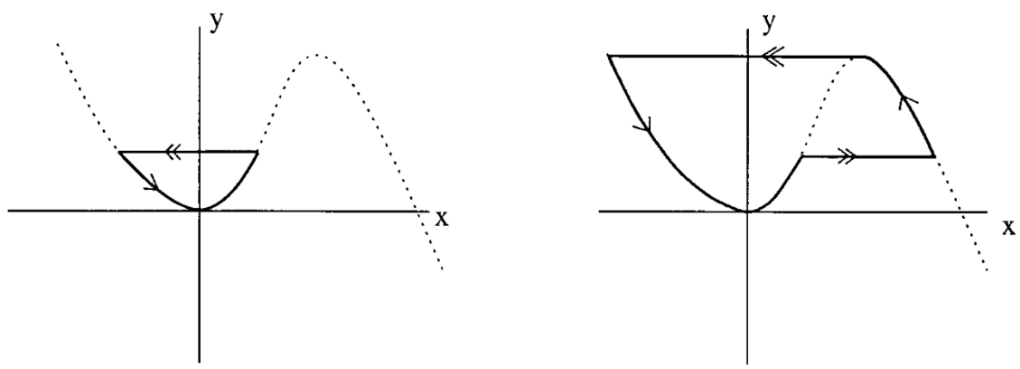

Figure 5: Singular canard cycles $\Gamma(s)$. Left: $s \in (0, y_M)$ (canard without head); right: $s \in (y_M, 2y_M)$ (canard with head).

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